

## Tab 2

# Keep On Track: Element B

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## Elements B:



### BIG 2D Marble Machine

Designed and crafted by Youtuber Jelle's Marble Runs, his video titled "BIG 2D Marble Machine in Continium Kerkrade" entailed a video demonstrating his creation. His marble wall is a series of indented intertubes that run along both horizontal and vertical paths— one filled with twists and turns that engage viewers. This machine is automatic meaning its a motor operating the chain shaft that carries up a marble at a time. When a marble reaches the top, it's dropped

down and luck guides it through one for four pathways available. This marble wall encapsulates exactly the kind of product our team wants to create for the lifeskills class at Whitmarsh Elementary.

The BIG 2D Marble Machine is an innovative piece of technology that demonstrates cause-and-effect. To the plinko-inspired mechanics to the long, exaggerated horizontal zigzags, kids with intellectual disabilities are provided with a visual representation of patterns and irregularities among functional objects.

A main set back to this marble wall is the limitation of possible motor skills that children have to master. There are no hands-on facets that allow kids with physical disabilities to manipulate the system while at the same time understanding the cause-and-effect dynamic.

The four unique paths the marble can possibly travel through is an idea that the group wants to replicate in our final design. The notion of chance and luck play a critical role in deciding the "fate" of the marble and that's the driving force behind the toy—to get kids to understand that each action has a reaction. Getting inspiration from the video, our team can transform this established concept to create a marble wall that will develop both motor skills and comprehension of cause-and-effect for kids with physical disabilities, while still catering to kids with intellectual disabilities.

### SOURCE



### Carousel Spinning Chair

The Carousel Spinning Chair is an interactive device that offers a chance for

kids to relieve some high energy through spinning around on the chair. Its tilted access means that very little energy needs to be inputted to achieve a powerful spinning motion that thoroughly stimulates the child using it.

This chair does extremely well with engaging the children using it to a very high degree. However, this engagement can only be held for a short period of time due to the nature of dizziness experienced by its users after a period of prolonged use. Additionally, it may not be as accessible to students who are not as advanced in the motor skills department. This means that while the device is sufficient in engaging certain students, it is just not capable of being accessed by the entire, varying demographic of a LifeSkills class. Keeping this in mind, different variations or modifications to promote universal accessibility will be necessary for a device that will be utilized by a varying population of students that all have different needs.

## SOURCE



### Crosswalk/Traffic Light

Plymouth Whitemarsh's Pre-AP Engineering course has worked alongside Whitemarsh Elementary's life skills class on numerous occasions. This partnership has created engaging prototypes, and the most impressive one as far is the traffic light and crosswalk simulation. Mixing in Arduino with CNC'd pieces, this group constructed a traffic light and crosswalk that helps children with physical and intellectual disabilities learn how to encounter real-life scenarios. In the safety of their classroom, the kids were physically taught what green, red, and yellow traffic lights signify all while being able to use the crosswalk feature to tie in pedestrian etiquette into the lesson. To take such a rudimentary ability and to flesh out the idea to be inclusive to those who are stunted in social cues really demonstrates a common good deed by creating prototypes for the lifeskills class.

On a surface level, the crosswalk and traffic light is a design rooted in coding and Arduino, and that's not a factor our group is very keen on. Our main incentive is practicality, something simple for the teachers. A low effort activity does not correlate with superficiality though, a marble wall is a difficult item to design but the manufacturing doesn't have to be above and beyond. When you have bits of Arduino it's a hefty reliability to put onto the teachers—who, in most cases, have no clue what Arduino is. So while our group is swaying away from the coding side of engineering, we are concentrating on developing motor skills for these children by including mechanical/movable bits that propel the marble along its path. By reviewing the portfolio for this product, our team came to a conclusion which targets more of the emotional side to kids rather than teaching them very critical life skills.

## SOURCE



### The Baby Einstein Magic Touch Piano

The Baby Einstein Magic Touch Piano has six interactive

songs that are for color-coordination and other simple fine motor abilities. When a child presses on a colored key that matches with the listed notes, the toy emits a sound that resembles actual piano notes. The volume adjusts to ensure that anybody, no matter their sensory limit, can enjoy this toy.

This machine, while marketed for babies, can develop crucial motor skills for children with either physical or intellectual disabilities. Its durability is crucial; The piano is made majority out of wood meaning it's less susceptible to breakage if it were made of plastic. The downside to it being wood is that it's very dangerous if a child were to throw the toy; although, that can be said about any toy. Otherwise, the colored coded keys and notes are perfect for developing coordination skills among children with disabilities. It teaches them cause and effect due to the music the toy plays each time a key is pressed. On a more personal level, it can be a creative outlet for a child and be the gateway to further musical pursuit.

### SOURCE



### Marble Machine

The Winter Garden Marble Machine is a specially-crafted music instrument that was designed and made by Youtuber “Wintergatan” in his video titled “Wintergatan - Marble Machine (music instrument using 2000 marbles)”. In the most simple way possible, this marble machine looks like the inside components of a piano, but if you made it run vertically. So in essence, it's similar to playing a piano but rather, the marble almost replicates the soundways that travel throughout the instrument to reach the mechanics which play out the music notes. This fun machine depicts the fun ways music and technology can coexist.

Music and movement go hand-in-hand to formulate a beautiful symphony. The marbles stimulating rhyme and music in general is the function that most appeals to our group as a sense of ease is critical in calming down kids with disabilities. While this instrument is very expert dependent, there are still main functions that can guide our brainstorming. Our rendition of a marble wall should not be complicated but not superficial so the incorporation of musical notes will act as another positive characteristic to help Whitemarsh’s life skills class.

What ties in cause and effect is the numerous parts that all cooperate in unison to generate music. The ability for marble to travel through this instrument is all due to the gears, knobs, innertubes, levers, shafts, and etc, working in harmony with each other. The unity of complex parts is symbolism of great craftsmanship and that is something our group wants to emulate in our design. The more that the different mechanical bits interact with each other, the more meaningful the cause-and-effect demonstrations will be. The main purpose for the marble wall has to be the lessons that the children are taking away, it's not all about the cool factor.

### SOURCE